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Problem Statement

The goal of this project is to analyze customer purchase behavior by identifying trends in sales revenue, product demand, and member loyalty engagement. By building a dashboard, we are able to provide actionable insights to increase revenue, reduce order cancellations, and improve overall customer satisfaction.

About Dataset

The data used to create this dashboard was extracted from the *Customer purchase behavior - Electronic Sales Data* dataset via Kaggle. All data was auto-generated for analysis purposes and does not represent real-world transactions or use actual customer data.This dataset contains exactly 20,000 rows of sales transaction data for an unspecified electronics company over a one-year period, from September 2023 to September 2024. There are no duplicate rows or columns. Key features to be used for the dashboard include: Customer ID, Loyalty Member, Product Type, Rating, Order Status, Total Price, Quantity, and Purchase Date.

Customer ID	Age	Gender	Loyalty Member	Product Type	SKU	Rating	Order Status	Payment Method	Total Price	Unit Price	Quantity	Purchase Date	Shipping Type	Add-ons Purchased	Add-on Total
16018	68	Female	No	Tablet	TBL345	4	Cancelled	Bank Transfer	3145.64	786.41	4	2024-01	Same Day	Impulse Item, Impulse Ite	46.8
16062	52	Male	No	Tablet	TBL345	5	Completed	Bank Transfer	3932.05	786.41	5	2024-02	Same Day	Impulse Item, Impulse Ite	134.71
16226	21	Female	No	Tablet	TBL345	1	Completed	Bank Transfer	786.41	786.41	1	2024-01	Same Day	Extended Warranty, Acce	168.8
16263	50	Female	No	Tablet	TBL345	1	Completed	Bank Transfer	4718.46	786.41	6	2024-02	Same Day	Impulse Item, Extended W	141.71
16294	66	Female	No	Tablet	TBL345	4	Completed	Bank Transfer	1572.82	786.41	2	2024-06	Same Day	Impulse Item, Impulse Ite	132.25
16341	79	Female	No	Tablet	TBL345	4	Completed	Bank Transfer	3932.05	786.41	5	2024-05	Same Day	Accessory	85.85
16357	23	Male	No	Tablet	TBL345	1	Cancelled	Bank Transfer	7077.69	786.41	9	2024-03	Same Day	Extended Warranty, Impu	142.44
16405	65	Female	No	Tablet	TBL345	2	Cancelled	Bank Transfer	1572.82	786.41	2	2024-01	Same Day	Extended Warranty	41.76
16418	76	Male	No	Tablet	TBL345	1	Cancelled	Bank Transfer	7077.69	786.41	9	2024-09	Same Day	Extended Warranty	66.74
16553	56	Female	Yes	Tablet	TBL345	4	Completed	Bank Transfer	1572.82	786.41	2	2024-05	Same Day	Impulse Item	20.08
16570	74	Male	No	Tablet	TBL345	1	Completed	Bank Transfer	3145.64	786.41	4	2024-03	Same Day	Accessory	50
16609	65	Male	Yes	Tablet	TBL345	4	Completed	Bank Transfer	3145.64	786.41	4	2024-09	Same Day	Impulse Item, Accessory	82.9
16901	67	Female	Yes	Tablet	TBL345	5	Completed	Bank Transfer	6291.28	786.41	8	2024-05	Same Day	Accessory	80.07
16976	26	Male	No	Tablet	TBL345	4	Completed	Bank Transfer	1572.82	786.41	2	2024-03	Same Day	Impulse Item	44.72
17054	52	Female	Yes	Tablet	TBL345	4	Completed	Bank Transfer	3932.05	786.41	5	2024-01	Same Day	Impulse Item, Impulse Ite	101.19
17075	65	Male	No	Tablet	TBL345	1	Cancelled	Bank Transfer	4718.46	786.41	6	2024-01	Same Day	Extended Warranty, Acce	135.47
17088	37	Male	No	Tablet	TBL345	5	Completed	Bank Transfer	5504.87	786.41	7	2024-05	Same Day	Impulse Item, Accessory,	259.34
17108	54	Male	Yes	Tablet	TBL345	4	Completed	Bank Transfer	3145.64	786.41	4	2024-07	Same Day	Extended Warranty	43.67
17257	67	Female	No	Tablet	TBL345	2	Completed	Bank Transfer	3932.05	786.41	5	2024-01	Same Day	Accessory, Impulse Item	69.85
17292	80	Female	No	Tablet	TBL345	5	Completed	Bank Transfer	4718.46	786.41	6	2024-08	Same Day	Extended Warranty, Exter	112.63
17306	68	Male	No	Tablet	TBL345	1	Completed	Bank Transfer	7077.69	786.41	9	2024-04	Same Day	Accessory	50.9
17314	77	Female	Yes	Tablet	TBL345	5	Completed	Bank Transfer	1572.82	786.41	2	2024-08	Same Day	Impulse Item, Impulse Ite	155.91
17470	30	Female	No	Tablet	TBL345	1	Completed	Bank Transfer	786.41	786.41	1	2024-05	Same Day	Accessory, Impulse Item	78.87
17471	69	Female	No	Tablet	TBL345	2	Completed	Bank Transfer	786.41	786.41	1	2024-06	Same Day	Extended Warranty, Exter	215.47
17596	79	Female	No	Tablet	TBL345	2	Completed	Bank Transfer	5504.87	786.41	7	2024-06	Same Day	Impulse Item	34.17
17690	18	Male	No	Tablet	TBL345	5	Completed	Bank Transfer	7864.1	786.41	10	2024-08	Same Day	Impulse Item	36.45
17865	48	Male	No	Tablet	TBL345	1	Completed	Bank Transfer	4718.46	786.41	6	2024-02	Same Day	Extended Warranty	16.88

Data Preparation

Key Transformations

All date columns were standardized and changed from Date/Time format to Date format for visual simplicity. Two new columns, such as “YearMonth” and “Customer Status”, were created to support the analysis.

The “YearMonth” column reformats the purchase date column into a “yyyy-MMM" structure, enabling cleaner axis labels for our visualizations and avoiding the default date hierarchy format.

The customer status column classifies a customer by their loyalty status, where they are categorized as a non member, an existing member, a new member, or a churned customer. The data was based on values from the “Loyalty Member” field. A customer is classified as “Churned” if they were a loyalty member in a previous purchase but not in their most recent purchase. Similarly, a customer is classified as a “New Member” if they were not a member previously, and became one in their most recent purchase.

New Measures and Calculations using DAX

Total Revenue: Calculated by summing the total price of orders with a status of “Completed”.

Average Order Value (AOV): Calculated by dividing the total revenue by the total number of orders (i.e. the total number of rows).

Units Sold: Calculated by summing the number of units purchased of the completed orders.

Customer Satisfactory Score (CSAT): Calculated by converting the average ratings from a 1 to 5 scale into a 0 to 1 scale, which was then changed to percentage format for proper display.

Electronic Sales KPI Dashboard Overview

Dashboard Highlights

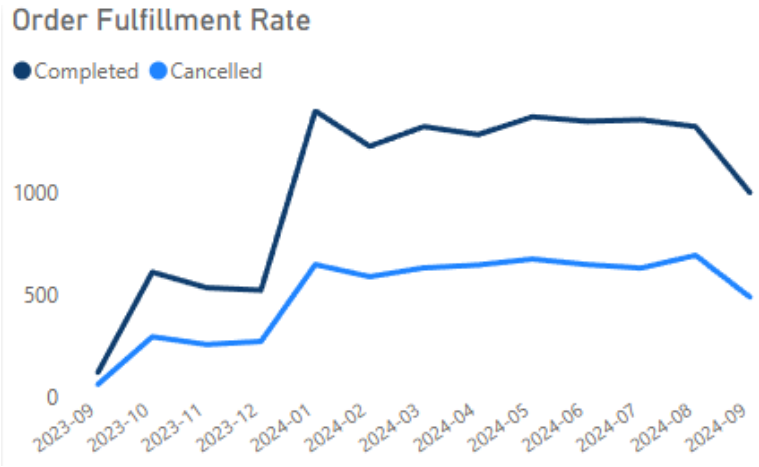
- Total Revenue made from completed orders **$42.63M**
- Total number of distinct customers **12,136**
- Average Order Value (AOV) **$2.13K**
- Total units sold **73.7K**
- Customer satisfaction score (CSAT) vs. hypothetical goal (80%) **52.35%**

Visualizations

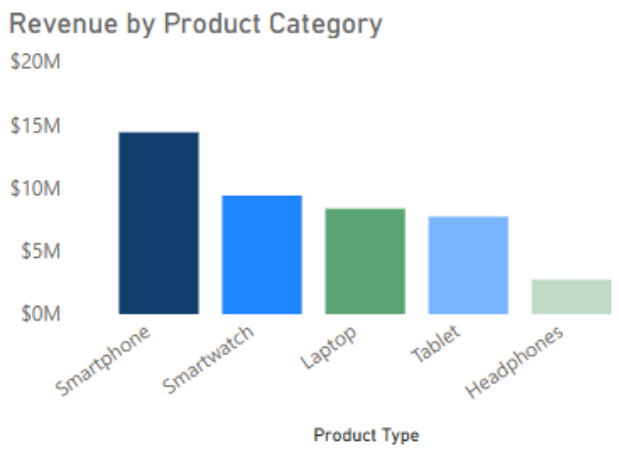
- Gauge meter to highlight the actual CSAT score to the hypothetical target
- Donut chart for comparison between the number of completed orders vs. canceled orders
- Donut chart for comparison of customer distribution by loyalty status
- Line chart comparing completed and cancelled orders over time
- Bar chart of revenue by product category
- Bar chart of revenue accumulated over time (MoM)

Key Performance Insights (KPI)

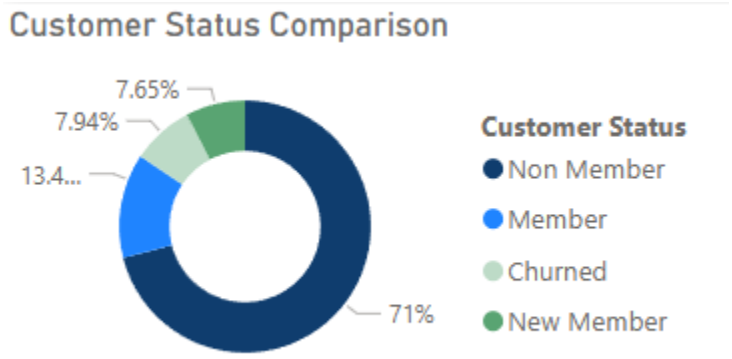
The total revenue generated from completed sales reached approximately $42.63 million. The average order value (AOV) and units sold show strong volume and high value per transaction.



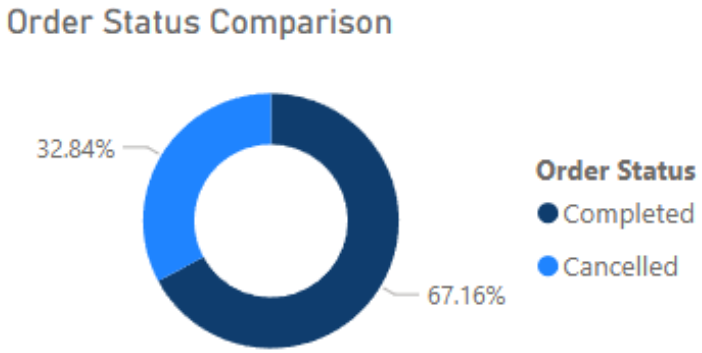
The number of orders peaked during January 2024, indicating strong seasonal demand at the beginning of the year. Meanwhile, order volume dipped near the end of the year showing lower seasonal demand.



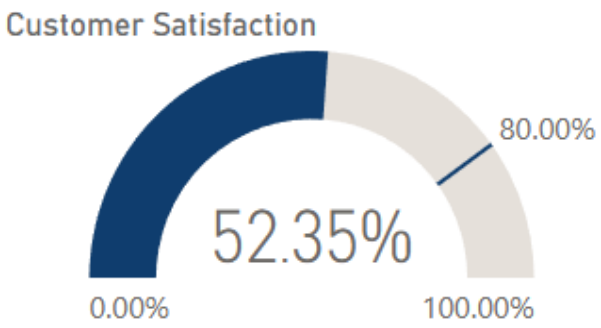
Smartphones were the most purchased product followed by smartwatches, indicating high market demand, while headphones were the least bought.



Significantly more purchases were made by non-members compared to loyalty members. This suggests that there are not enough benefits for loyalty members. This suggests that the current loyalty program does not provide enough benefits for loyalty members.



The cancellation rate for orders is 32.84%, which is concerning and may indicate issues in the payment or fulfillment process.



The average customer satisfaction score is 52.35%, implying that roughly half the customers are unhappy or neutral about their purchase.

Recommendations

- Revamp loyalty program and introduce stronger incentives to encourage higher spending and engagement by members
- Investigate the reasons for high cancellation rates and strategize reduction methods
- Enhance customer support and satisfaction by offering multi-channel support, reviewing surveys/feedback, reducing response times for complaints and inquiries, and enhancing product quality control.

Further Considerations

Seasonal Trends

- Analyze the connection between order status and seasonal trends (Autumn, Winter, Spring, and Summer)

Customer Demographics

- Analyze purchase behavior by gender
- Analyze purchase behavior by age group

Order Fulfillment

- Analyze the connection between add-ons and shipping impact
- Investigate if the shipping type affects the cancellation rate of orders
- Investigate which transaction methods are generally preferred by customers

Github Link: https://github.com/melzhen/Electronic-Sales-Dashboard

Reference

Seamons, C. (2023). *Customer purchase behavior - Electronic Sales Data*. Kaggle.com. https://www.kaggle.com/datasets/cameronseamons/electronic-sales-sep2023-sep2024